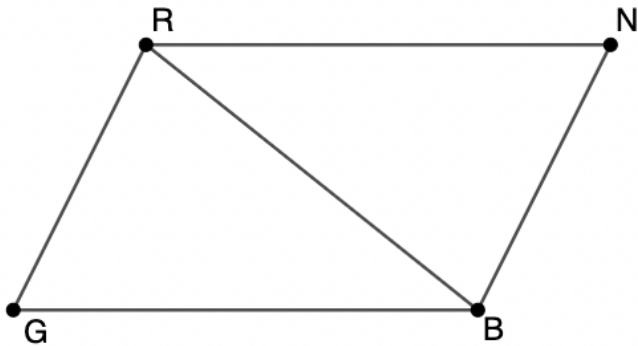


Geometry  
Unit 8  
Lesson 1 Homework

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Quadrilateral  $RNBG$  is shown with diagonal  $\overline{RB}$ .



**Given:**  $\overline{RN} \cong \overline{GB}$ ,  $\angle NRB \cong \angle GBR$

**Prove:**  $RNBG$  is a parallelogram.

| Statement                                  | Reason                                                                                                                      |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| $\overline{RN} \cong \overline{GB}$        | Given                                                                                                                       |
| 1. $\overline{RB} \cong \overline{RB}$     | 2. Reflexive Property of Congruence                                                                                         |
| $\angle NRB \cong \angle GBR$              | Given                                                                                                                       |
| $\triangle RGB \cong \triangle RNB$        | 3. SAS $\cong$                                                                                                              |
| 4. $\overline{RN} \parallel \overline{GB}$ | If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel. |
| 5. $\angle BRG \cong \angle RBN$           | 6. Corresponding Parts of Congruent Triangles are Congruent                                                                 |
| $\overline{RG} \parallel \overline{NB}$    | If two lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel. |
| $ABCD$ is a parallelogram                  | 7. A parallelogram has two sets of opposite parallel sides.                                                                 |

Use the statements and reasons provided to complete the two-column proof.

|                                         |                                                           |
|-----------------------------------------|-----------------------------------------------------------|
| $\angle BRG \cong \angle RBN$           | Corresponding Parts of Congruent Triangles are Congruent  |
| $\angle RGB \cong \angle NBR$           | Reflexive Property of Congruence                          |
| $\overline{RB} \cong \overline{RB}$     | Transitive Property of Congruence                         |
| $\overline{RG} \cong \overline{NB}$     | SSS $\cong$                                               |
| $\overline{RN} \parallel \overline{GB}$ | SAS $\cong$                                               |
| $\overline{RG} \parallel \overline{NB}$ | A parallelogram has two sets of opposite parallel sides.  |
|                                         | A parallelogram has two sets of opposite congruent sides. |

Fill in the blanks with the correct terms to complete the statements.

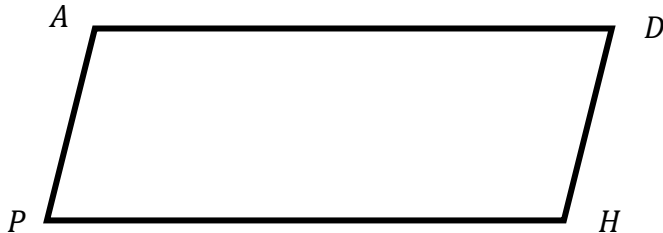
8. A quadrilateral containing two set(s) of parallel/congruent sides is  
a parallelogram.

9. Opposite interior angles of a parallelogram are congruent.

10. The diagonals of a parallelogram bisect each other.

**Use the given information to answer questions 1-3.**

Quadrilateral  $ADHP$  is shown with  $HP = 25$ ,  $AD = 8x + 21$ ,  $DH = 20x - 4$  where  $x = \frac{1}{2}$ .



1. I can conclude ...

**that quadrilateral  $ADHP$  is a parallelogram.**

2. I know this because ...

**there are two pairs of opposite sides of equal length.**

3. If  $ADHP$  is a parallelogram, what is the length of  $\overline{PA}$ ?

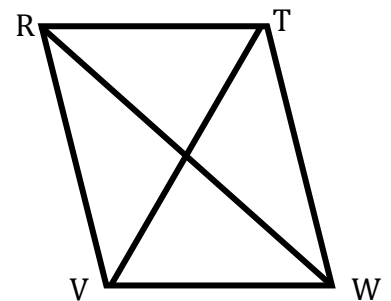
**$PA = 6$**

**Use the given information to answer questions 4-6.**

Quadrilateral  $RTWV$  is shown with diagonals  $\overline{RW}$  and  $\overline{TV}$  that intersect at point  $M$  where  $RM = 4x$ ,  $TM = (3y + 15)$ ,  $WM = 84$ , and  $VM = 75$ .

4. What value of  $x$  will make  $RTWV$  a parallelogram?

**$4x = 84$   
 $x = 21$**



5. What value of  $y$  will make  $RTWV$  a parallelogram?

$$3y + 15 = 75$$

$$3y = 60$$

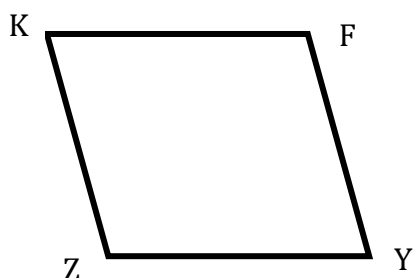
$$y = 20$$

6. What is the length of  $\overline{VT}$ ?

$$150$$

**Use the given information to answer questions 7-10.**

Quadrilateral  $KFYZ$  is shown where  $m\angle F = (20z - 4)^\circ$ ,  $m\angle Y = 54^\circ$ , and  $m\angle Z = (8t)^\circ$ , where  $t = 15\frac{3}{4}$ .



Devonte and Briyel are discussing whether  $KFYZ$  is a parallelogram. Briyel says it is, and Devonte says it is not.

7. Who is correct?

**Briyel is correct.**

8. Which property of parallelograms was used to determine if  $KFYZ$  is a parallelogram?

**Consecutive interior angles are supplementary.**

9. What is the value of  $z$ ?

$$20z - 4 = 126$$

$$20z = 130$$

$$z = 6.5$$

10. What is the  $m\angle K$ ?

$$m\angle K = 54^\circ$$

Geometry

Unit 8

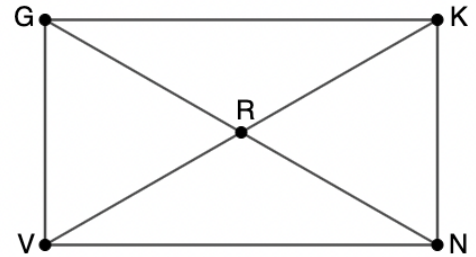
Lesson 3 Homework

Name **Answer Key** \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

Rectangle  $GKNV$  is shown with diagonals  $\overline{GN}$  and  $\overline{KV}$  intersecting at  $R$  where  $GK = 26x - 13$  and  $NV = 14x + 5$ .



1. Select all true properties of  $GKNV$ .
  - ☐ Opposite sides are parallel.
  - ☐ Opposite sides are congruent.
  - ☐ Consecutive angles are congruent.
  - ☐ All angles are right angles.
  - ☐ Diagonals are perpendicular.
  - ☐ All sides are congruent.

2. What is the value of  $x$ ?

$$x = 1.5$$

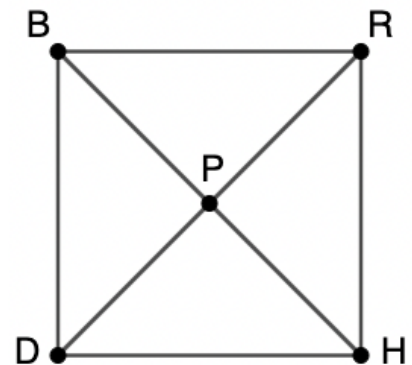
3. What is the length of  $\overline{NV}$ ?

$$NV = 26$$

Square  $BRHD$  with diagonals  $\overline{BH}$  and  $\overline{RD}$  intersecting at point  $P$ .

4. Write two congruent statements using the diagonals of  $BRHD$ .

$$\begin{aligned} \overline{BH} &\cong \overline{RD} \\ \overline{BP} &\cong \overline{HP} \\ \overline{DP} &\cong \overline{RP} \\ \overline{BP} &\cong \overline{HP} \cong \overline{DP} \cong \overline{RP} \end{aligned}$$



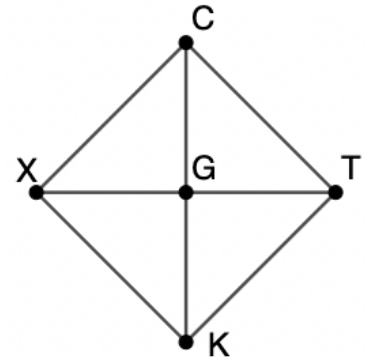
5. Given  $BP = 32y - 10$ , and  $HP = 18y + 32$ . What is the value of  $y$ ?

$$y = 3$$

6. What is the length of  $\overline{RD}$ ?

$$RD = 172$$

Quadrilateral  $CTKX$  is shown with diagonals  $\overline{CK}$  and  $\overline{XT}$  intersecting at point  $G$ .



7. What value of  $g$  makes  $CTKX$  a rhombus, if  $CT = 55$ , and  $TK = 15g + 19$ ?

$$g = 2.4$$

8. Given that  $CTKX$  is a rhombus with  $CG = 6y - 17.81$  and  $KG = 3y + 10.54$ . What is the length of  $\overline{TX}$ ?

$$y = 9.45$$

$$TX = 77.78$$

9. What properties are the same between  $GKNV$  and  $CTKX$ ?

Opposite sides are congruent

Opposite angles are congruent

Consecutive angles are supplementary

Opposite sides are parallel

Diagonals bisect each other

10. What properties are different between  $BRHD$  and  $CTKX$ ?

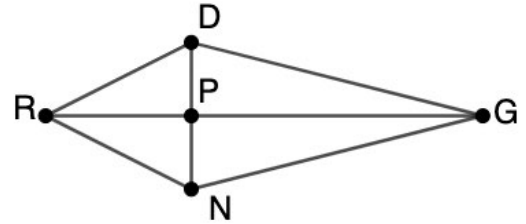
Diagonals are congruent to each other in  $BRHD$

Quadrilateral  $BRHD$  has four right angles

Geometry  
Unit 8  
Lesson 4 Homework

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Kite RDGN is shown with diagonals  $\overline{RG}$  and  $\overline{DN}$  intersecting at point P where  $DP = 13$  and  $RP = 15$ .



1. What is the length of  $\overline{RD}$ ?

$$RD^2 = 13^2 + 15^2$$

$$RD^2 = 169 + 225$$

$$RD = \sqrt{394} \approx 19.849$$

2. Given  $PG = 22$ . What is the length of  $\overline{NG}$ ?

$$NG^2 = 13^2 + 22^2$$

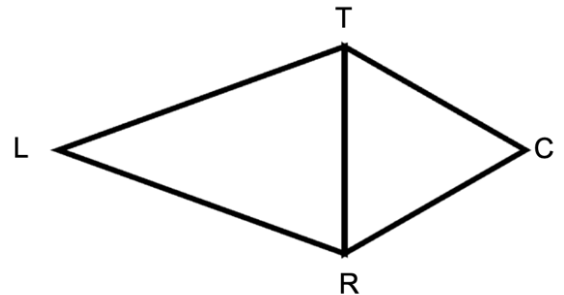
$$NG^2 = 169 + 484$$

$$NG = \sqrt{653} \approx 25.554$$

3. What is the perimeter of RDGN?

$$\text{Perimeter} = 2 \cdot 19.849 + 2 \cdot 25.554 = 90.806$$

TCRL is a kite as shown where  $m\angle CTR = (7x - 12)^\circ$ ,  $m\angle TCR = (8x + 32)^\circ$ ,  $m\angle LRT = (9y + 40)^\circ$ , and  $m\angle TLR = (4y)^\circ$ .



4. What is the value of  $x$ ? Round your answer to the nearest hundredth.

$$8x + 32 + 2(7x - 12) = 180$$

$$8x + 32 + 14x - 24 = 180$$

$$22x + 8 = 180$$

$$x = 7.82$$

5. What is the value of  $y$ ? Round your answer to the nearest hundredth.

$$4y + 2(9y + 40) = 180$$

$$4y + 18y + 80 = 180$$

$$22y = 100$$

$$y = 4.55$$

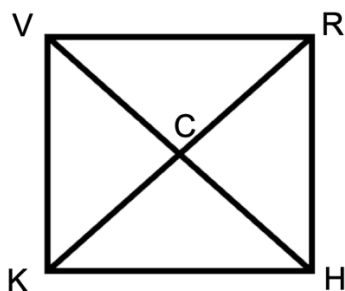
6. What is the measure of  $\angle CTL$ ?

$$m\angle CTL = m\angle CTR + m\angle RTL$$

$$m\angle CTL = 42.74 + 80.95$$

$$m\angle CTL = 123.69^\circ$$

Quadrilateral VRHK is shown with diagonals  $\overline{VH}$  and  $\overline{RK}$  intersecting at point C.



7. What must be true for VRHK to be a square?

All sides are congruent  
All angles are congruent  
Diagonals are congruent  
Diagonals bisect each other

8. Given VRHK is a square with  $VC = 6z$ , and  $VR = 12$ . What is the value of  $z$ ? Leave your answer in radical form.

$$\begin{aligned}(6z)^2 + (6z)^2 &= 12^2 \\ 36z^2 + 36z^2 &= 144 \\ 72z^2 &= 144 \\ z &= \sqrt{2}\end{aligned}$$

9. What is the length of  $\overline{RK}$ , in simplest radical form?

$$RK = 6\sqrt{2} + 6\sqrt{2} = 12\sqrt{2}$$

10. What is the length of  $\overline{VK}$ , in simplest radical form?

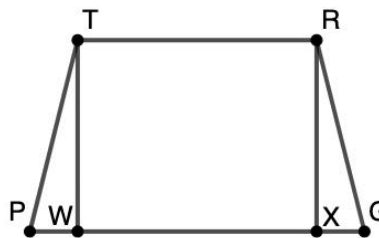
$$VK = 12$$

Geometry  
Unit 8  
Lesson 5 Homework

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

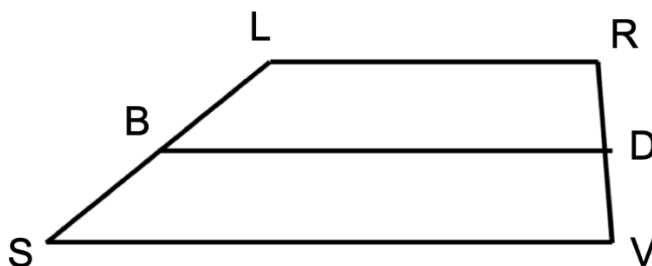
Given:  $TRGP$  is an isosceles trapezoid with altitudes  $\overline{TW}$  and  $\overline{RX}$ .

Prove:  $\overline{PW} \cong \overline{XG}$



| Statement                                                                   | Reason                                                   |
|-----------------------------------------------------------------------------|----------------------------------------------------------|
| $TRGP$ is an isosceles trapezoid                                            | Given                                                    |
| 1. $\overline{TW} \cong \overline{RX}$                                      | Altitudes of an isosceles trapezoid are congruent.       |
| $\overline{TW} \perp \overline{PG}$ and $\overline{RX} \perp \overline{PG}$ | 2. Definition of altitude                                |
| $\angle TWP$ and $\angle RXG$ are right angles.                             | Perpendicular lines form right angles                    |
| 3. $\angle TWP \cong \angle RXG$                                            | All right angles are congruent                           |
| 4. $\angle TPW \cong \angle RGX$                                            | 5. Base angles of an isosceles trapezoid are congruent   |
| 6. $\triangle TPW \cong \triangle RGX$                                      | AAS Congruence Theorem                                   |
| $\overline{PW} \cong \overline{XG}$                                         | Corresponding parts of congruent triangles are congruent |

Trapezoid  $LRVS$  is shown with the midsegment  $\overline{BD}$  where  $LB = 5 + 3x$ ,  $SB = 5x - 11$ ,  $RD = 9y + 16$ ,  $VD = 51 + 4y$ .



7. What is the value of  $x$  and  $y$ ?

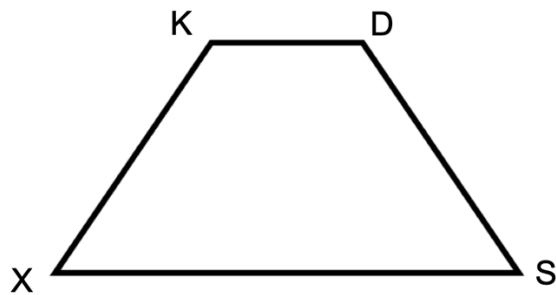
$$x = 8$$

$$y = 7$$

8. Find the length of  $\overline{LB}$  and  $\overline{RV}$ .

$$\begin{aligned} LB &= 29 \\ RV &= 158 \end{aligned}$$

$KDSX$  is an isosceles trapezoid, as shown, where  $m\angle K = (35z - 20)^\circ$ ,  $m\angle X = (15z + 8)^\circ$ ,  $m\angle S = (10t + 25)^\circ$ .



9. What is the value of  $z$  and  $t$ ? Round your answer to the nearest hundredth.

$$\begin{aligned} z &= 3.84 \\ t &= 4.06 \end{aligned}$$

10. What is  $m\angle S$  and  $m\angle D$ ? Round your answer to the nearest tenth.

$$\begin{aligned} m\angle S &= 65.6^\circ \\ m\angle D &= 114.4^\circ \end{aligned}$$

Geometry

Unit 8

Lesson 6 Homework

Name **Answer Key** \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

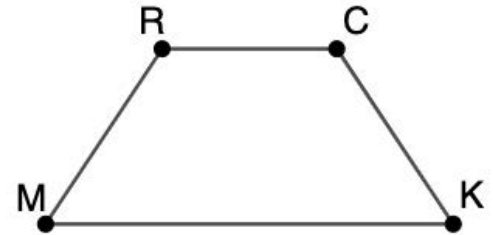
Two congruent angles isosceles trapezoid  $KMRC$  have measures  $(3x + 10)^\circ$  and  $(5x - 10)^\circ$ .

1. What is the value of  $x$ ?

$$3x + 10 = 5x - 10$$

$$20 = 2x$$

$$x = 10$$



2. What are the measures of all angles of the trapezoid?

$$m\angle M = m\angle K = 40^\circ$$

$$m\angle R = m\angle C = 140^\circ$$

For each quadrilateral description, select all shapes that match the description.

3. Quadrilateral  $GRMW$  has two pairs of parallel sides and diagonals are congruent.

☐ Parallelogram

☒ Rectangle

☒ Square

☐ Rhombus

☐ Kite

4. Quadrilateral  $BNTV$  has diagonals are perpendicular to each other and two pairs of parallel sides.

☐ Parallelogram

☐ Rectangle

☒ Square

☒ Rhombus

☐ Kite

5. Quadrilateral  $KPSX$  has two pairs of congruent sides and diagonals are perpendicular to each other.

☐ Parallelogram

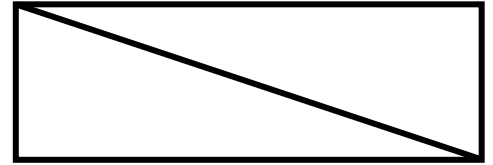
☐ Rectangle

☒ Square

☒ Rhombus

☒ Kite

Barrett is looking at a building on the school grounds which is shaped like a rectangle as shown. He knows the building has a height of  $25x$  feet and a width of  $28x$  feet with a diagonal of 180 feet.



6. What is the value of  $x$ ? Round your answer to the nearest hundredth.

$$\begin{aligned}(25x)^2 + (28x)^2 &= 180^2 \\ 1409x^2 &= 32400 \\ x^2 &= 22.995 \\ x &\approx 4.80\end{aligned}$$

7. What is the height of the building? Round your answer to the nearest hundredth.

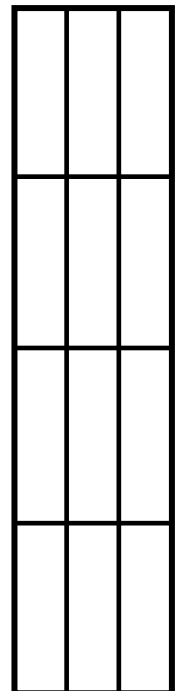
$$\text{height} \approx 25(4.80) \approx 120 \text{ feet}$$

8. What is the width of the building? Round your answer to the nearest hundredth.

$$\text{width} \approx 28(4.80) \approx 134.4 \text{ feet}$$

**Use the given figure to answer questions 9-10.**

A wall has paneling made up of rectangles, with an area of 30 square feet each, as shown. There are four rectangles along the height and three along the top. The edge of each rectangle in the paneling is  $(2x + 3)$  feet in width and  $(4x + 2)$  feet in height.



9. What is the value of  $x$ ?

$$\begin{aligned}30 &= (2x + 3)(4x + 2) \\ 0 &= 8(x + 3)(x - 1) \\ x &= -3 \text{ and } x = 1\end{aligned}$$

10. What are the dimensions of the portion of the wall covered the paneling?

$$\begin{aligned}\text{height of 1 panel} &= 4(1) + 2 = 6 \text{ feet} \\ \text{height of 4 panels} &= 4(6) = 24 \text{ feet}\end{aligned}$$

$$\begin{aligned}\text{width of 1 panel} &= 2(1) + 3 = 5 \text{ feet} \\ \text{width of 3 panels} &= 3(5) = 15 \text{ feet}\end{aligned}$$

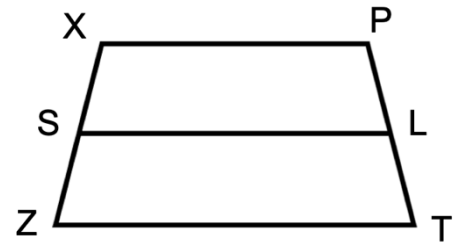
Geometry  
Unit 8  
Lesson 7 Homework

Name **Answer Key** \_\_\_\_\_  
Date \_\_\_\_\_  
Period \_\_\_\_\_

XZTP is a trapezoid where  $\overline{XP} \parallel \overline{ZT}$ .  $\overline{SL}$  is a midsegment of XZTP and  $XS = 7$ ,  $LT = 6$ ,  $SL = 10$ ,  $TZ = 2x - 4$ , and  $XP = x - 3$ .

1. Write an equation to show the relationship between  $\overline{SL}$  and the bases of the trapezoid.

$$SL = \frac{XP + ZT}{2}$$



2. What is the value of  $x$ ?

$$\begin{aligned} 10 &= \frac{2x - 4 + x - 3}{2} \\ 20 &= 2x - 4 + x - 3 \\ 27 &= 3x \\ x &= 9 \end{aligned}$$

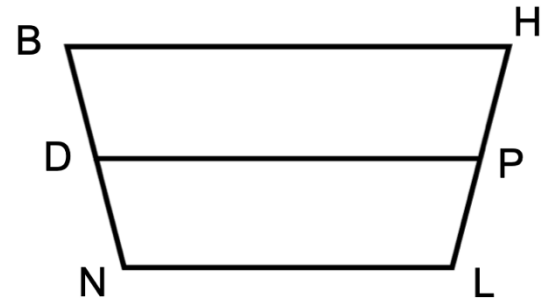
3. What is the length of  $\overline{XP}$  and  $\overline{ZT}$ ?

$$\begin{aligned} ZT &= 14 \\ XP &= 6 \end{aligned}$$

Trapezoid BHLN is shown where  $\overline{BH} \parallel \overline{NL}$  with midsegment  $\overline{DP}$  and  $BH = 5x + 12$ ,  $NL = 3x$ , and  $DP = 5x$ .

4. Find the value of  $x$ .

$$\begin{aligned} 5x &= \frac{5x + 12 + 3x}{2} \\ 10x &= 8x + 12 \\ 2x &= 12 \\ x &= 6 \end{aligned}$$



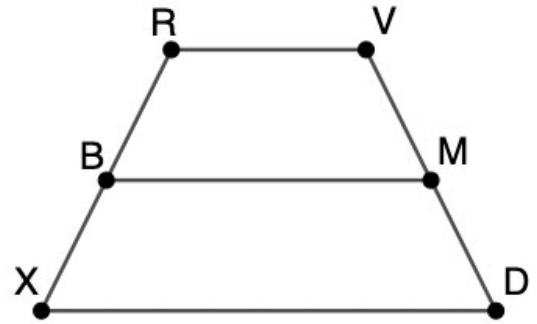
5. Find the length of  $\overline{DP}$ .

$$DP = 30$$

6. Find the lengths of  $\overline{BH}$  and  $\overline{NL}$ .

$$\begin{aligned} BH &= 42 \\ NL &= 18 \end{aligned}$$

Isosceles trapezoid  $RVDX$  is shown with midsegment  $\overline{BM}$  where  $RV = 2x + 1$ ,  $XD = 3x + 2$ ,  $BM = 2x + 4$  and  $MD = 3$ .



7. What is the value of  $x$ ?

$$\begin{aligned} 2x + 4 &= \frac{2x + 1 + 3x + 2}{2} \\ 4x + 8 &= 5x + 13 \\ x &= 5 \end{aligned}$$

8. What is the length of  $\overline{BM}$ ?

$$BM = 14$$

9. What is the length of  $\overline{XD}$  and  $\overline{RV}$ ?

$$\begin{aligned} XD &= 17 \\ RV &= 11 \end{aligned}$$

10. What is the perimeter of  $RVDZ$ ?

$$\begin{aligned} \text{Perimeter} &= RV + VD + XD + RX \\ \text{Perimeter} &= 11 + 2(3) + 17 + 2(3) \\ \text{Perimeter} &= 42 \end{aligned}$$